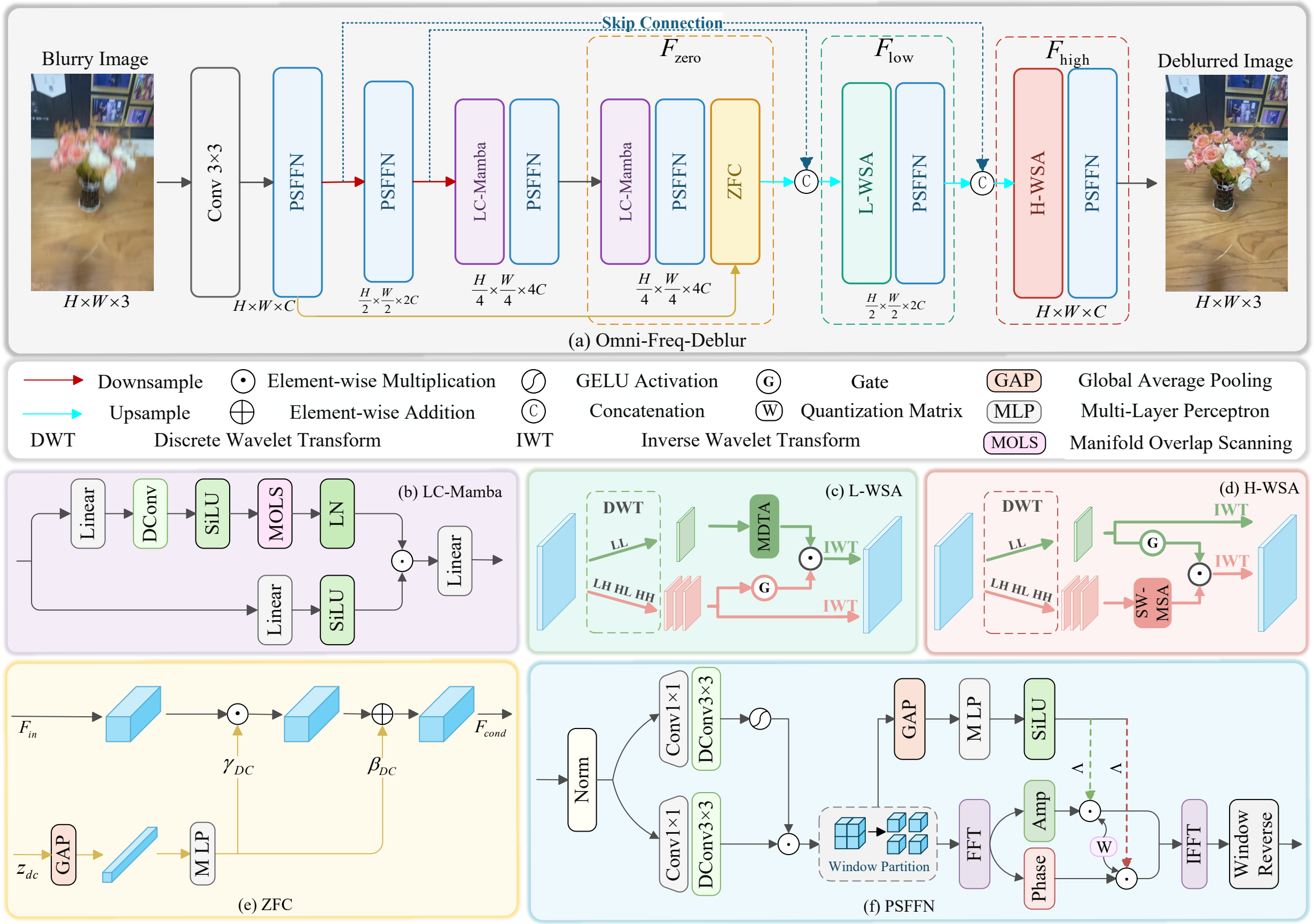
Skip Connection
Blurry Image
Conv 3×3
PSFFN
LC-Mamba
ZFC
L-WSA
H-WSA
Deblurred Image
F_{zero}
F_{low}
F_{high}
$H\times W\times 3$
$H\times W\times C$
$\frac{H}{2}\times\frac{W}{2}\times 2C$
$\frac{H}{4}\times\frac{W}{4}\times 4C$
(a) Omni-Freq-Deblur
Downsample
Upsample
Element-wise Multiplication
Element-wise Addition
GELU Activation
Concatenation
Gate
Quantization Matrix
Global Average Pooling
Multi-Layer Perceptron
DWT Discrete Wavelet Transform
IWT Inverse Wavelet Transform
MOLS Manifold Overlap Scanning
Linear
DConv
SiLU
MOLS
LN
(b) LC-Mamba
DWT
LL
LH HL HH
MDTA
IWT
G
(c) L-WSA
SW-MSA
(d) H-WSA
F_{in}
F_{cond}
γ_{DC}
β_{DC}
z_{dc}
GAP
MLP
(e) ZFC
Norm
Conv1×1
DConv3×3
Window Partition
FFT
Amp
Phase
W
IFFT
Window Reverse
(f) PSFFN